



SMART HEALTH CARE DIAGNOSIS

using Machine Learning





TEAM DETAILS:

S.no	STUDENT NAME	REG.NO
1	B.AKHILA SRI	99220040458
2	A.SURYA PRAKASH REDDY	99220040385
3	P.SHANMUKHA	99220040336
4	M.GOKUL REDDY	99220040309



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A" Grade



PROBLEM STATEMENT:

Developing a smart disease prediction system using machine learning and patient-reported symptoms to aid accurate diagnosis, intervention, and treatment planning. The solution includes data collection, feature extraction, model training, and validation for robust and reliable predictions across diverse diseases. Ensuring user-friendly accessibility, the system caters to varying symptom severity and complexity, revolutionizing healthcare delivery by providing timely and informed insights to healthcare professionals and patients alike.



WHY WE HAVE CHOOSSEN ??

- It has been proved that every month over 65% of the population in India has a tendency towards general malady which includes cough,cold,fever,Headaches .
- But lots of people don't realize that these symptoms can be more dangerous for them in future, even lack of ignorance may result into death.
- Hence the identification of these disease at early stage is very necessary .
- So, the purpose of our project is to make predictions for the commonly occurring symptoms that were unchecked and later become the fatal disease. At present, when one suffers from a particular disease, then the person has to visit a doctor which is time consuming and costly too and availability of doctor will be may or may not.
- So if use this system the patient saves their time and money and also completes their health diagnosis process very easily.



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A" Grade



INTRODUCTION:

- In a rapid evolving world technology is transforming in every aspect of our lives including health care
- This project endeavors to develop a sophisticated smart disease prediction system that utilizes machine learning algorithms. By analyzing patient-reported symptoms, this system aims to revolutionize diagnostic accuracy and timeliness.
- This project focuses on harnessing the power of machine learning models such as K-Nearest Neighbors (KNN), Naive Bayes, Random Forest, and Decision Trees for accurate and efficient disease diagnosis. By integrating these algorithms with patient symptom data, our objective is to develop a comprehensive smart disease prediction system capable of providing timely and precise insights to healthcare professionals.



OBJECTIVES:

- Early Disease Detection
- Improved Patient Outcomes
- Remote Monitoring
- Healthcare Accessibility
- Public Health Surveillance
- Patient Empowerment
- Automated diagnosis process
- Secure and compliant





CHALLENGES:

- **Data Quality and Availability:** Obtaining high-quality and comprehensive symptom data from patients can be challenging
- **Algorithm Complexity and Interpretability:** algorithms used for disease prediction can be complex and difficult to interpret
- **Overfitting and Generalization:** Ensuring that the machine learning models do not overfit to the training data and can generalize well to unseen data is crucial for reliable predictions.
- **Ethical and Privacy Concerns:** Ensuring or keeping the patient data private
- **Integration with Existing Healthcare Systems:** Integrating the smart disease prediction system with existing healthcare IT infrastructure
- **Algorithm Bias and Fairness:** Addressing potential biases in the machine learning algorithms essential to ensure equitable and fair
- **Interpretation and Communication of Results:** Effectively communicating the predictions generated by the system to healthcare professionals and patients in a clear and understandable manner presents challenges



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A" Grade



WORK FLOW:

- ✓ Problem Analysis
- ✓ Data Collection and Preparation
- ✓ Model Development and Training
- ✓ Model Evaluation and Optimization
- ✓ Deployment and Maintenance



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A" Grade



DATASET AND MODEL DESCRIPTION:

Dataset utilized in this method is during a structured format. The dataset that is employed contains the illness name with its all symptoms. As our system relies on supervised learning machine algorithms, the dataset has the label with zero or one. Then we tend to divide the dataset into a training dataset and Testing dataset. The model is trained by a training dataset contains 4920 rows. All algorithms were applied to the present training dataset, so the machine learning model is trained. Then the testing dataset was provided to the trained model to check the accuracy of the model.



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A" Grade



TECHNICAL STACK USED:

In this project standard libraries for information analysis and model creation are used.

The following libraries are utilized in this project:

- **tkinter**
- **Numpy**
- **Pandas**
- **sklearn** for the implementation of our model.
- **GUI** - GUI created for this project is a simple tkinter user interface consisting of labels, messagebox, button, text, title and possibility menu. Buttons are used to provide functionalities and predict the outcome of models additionally 2 utility buttons specifically exit and rest also are created.



METHODOLOGY:

- In this project, we elect to figure with some machine learning algorithms and which are:
- **Decision Tree algorithmic program** – we have a tendency to used decision tree to divide up big assortment of data/records with success into smaller sets of data/records by applying sequence of easy decision tree.
- **Naïve Byes algorithmic program** – this can be a simple and powerful prophetic modelling. This essentially belongs to the family of Probability classifiers supported applying the mathematician bayes theorem.
- **K Nearest Neighbor algorithmic program** – it's a supervised learning that is employed classification and regression.
- **Random Forest algorithmic program** – it's a supervised learning, and it makes decision trees on the various information collections/samples so takes majority vote for the classification and the averages of them just in case of regression.

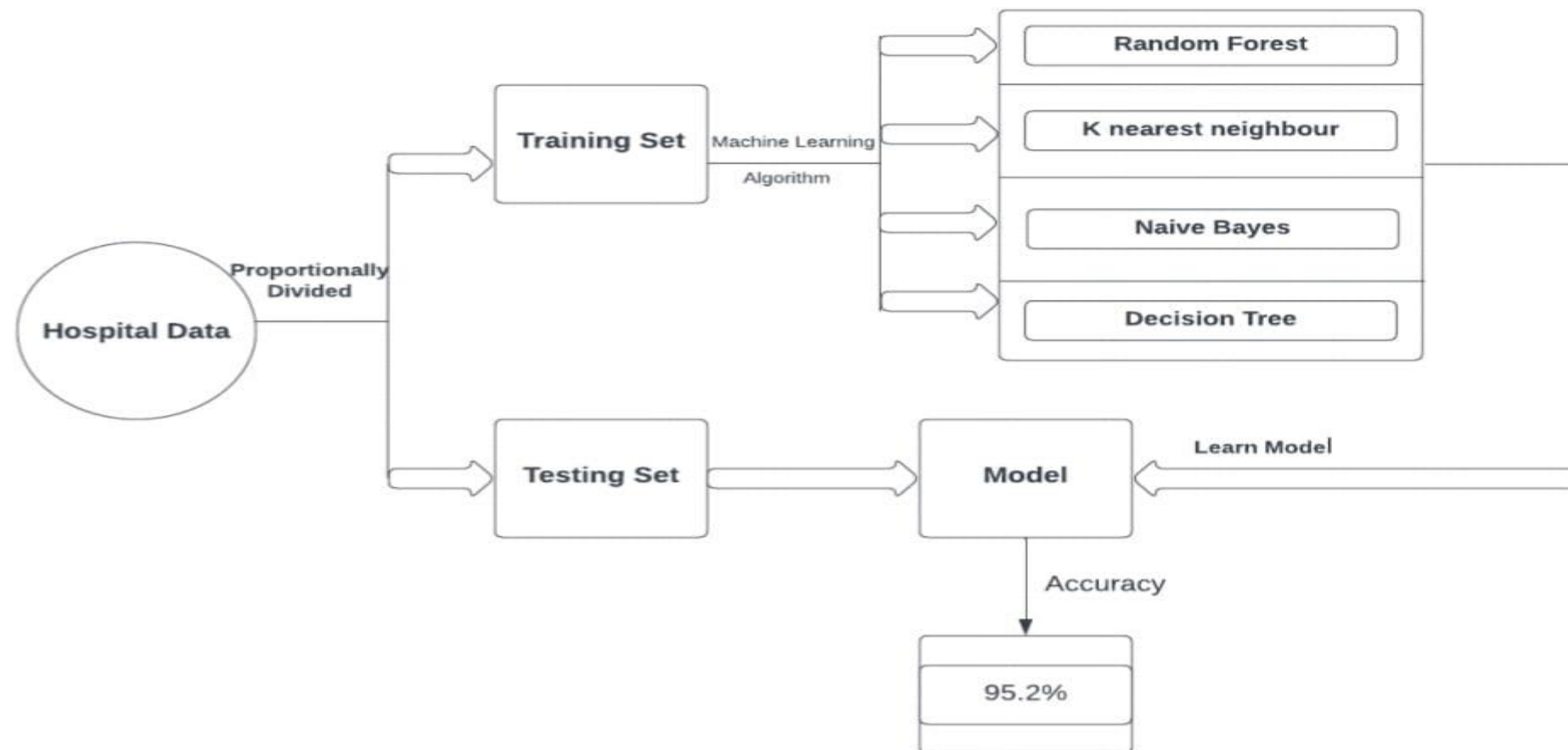


SYSTEM DESIGN:

- The overall disease prediction system performs supervised learning algorithm which is the crux of the system.
- The user needs to enter the name first then the five symptoms.
- Disease will be thrown as output based on those symptoms; List of symptoms is divided into levels based on the nature of disease.
- Model is revolved around four different algorithms Decision tree, Random Forest, KNN and Naïve bayes.
- Algorithm will be called when user hit the predict button, predict 1, 2, 3 and 4 button is for Decision tree, Random Forest, KNN, Naive byes respectively.
- There should be at least two symptoms required to get the prediction, Confusion matrix and accuracy score predicts the output of our model for each algorithm and separate table is created which stores the input and output data. Accuracies given by decision tree, Random Forest, KNN and Naive bayes are 95.6%,93.66%,92% and 95.1%



BLOCK DIAGRAM:

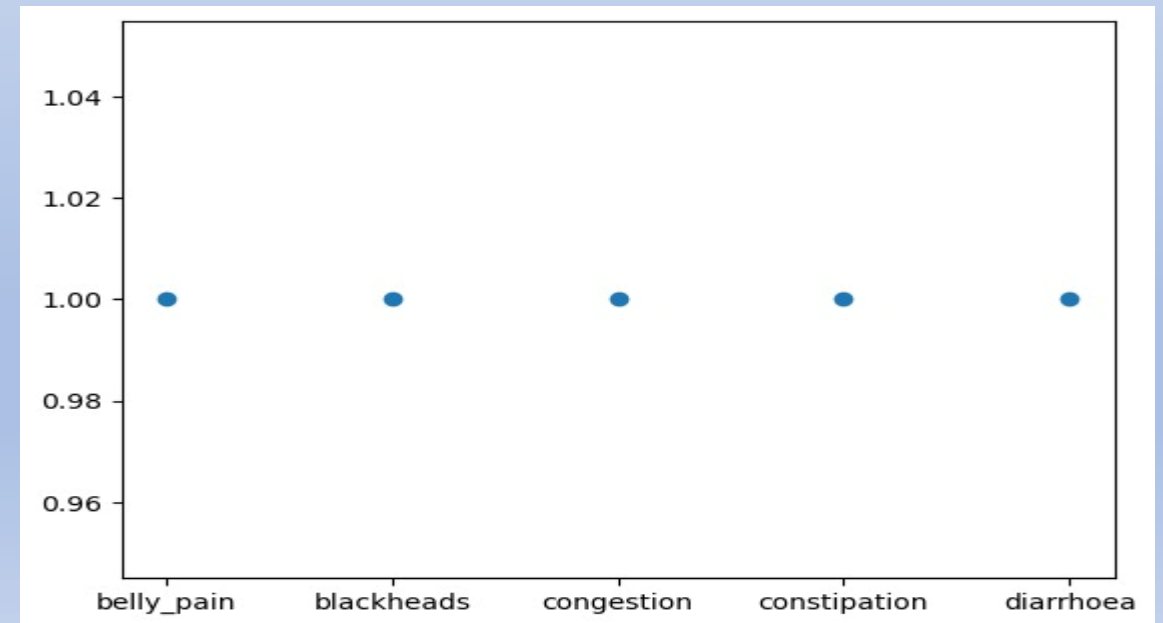




IMPLEMENTATION:

Decision tree: A decision tree is a supervised machine learning algorithm used for classification and regression tasks. It recursively splits the data into subsets based on the features' values, aiming to create a tree-like model that predicts the target variable by following a path of binary decisions. It's intuitive, interpretable, and can handle both numerical and categorical data.

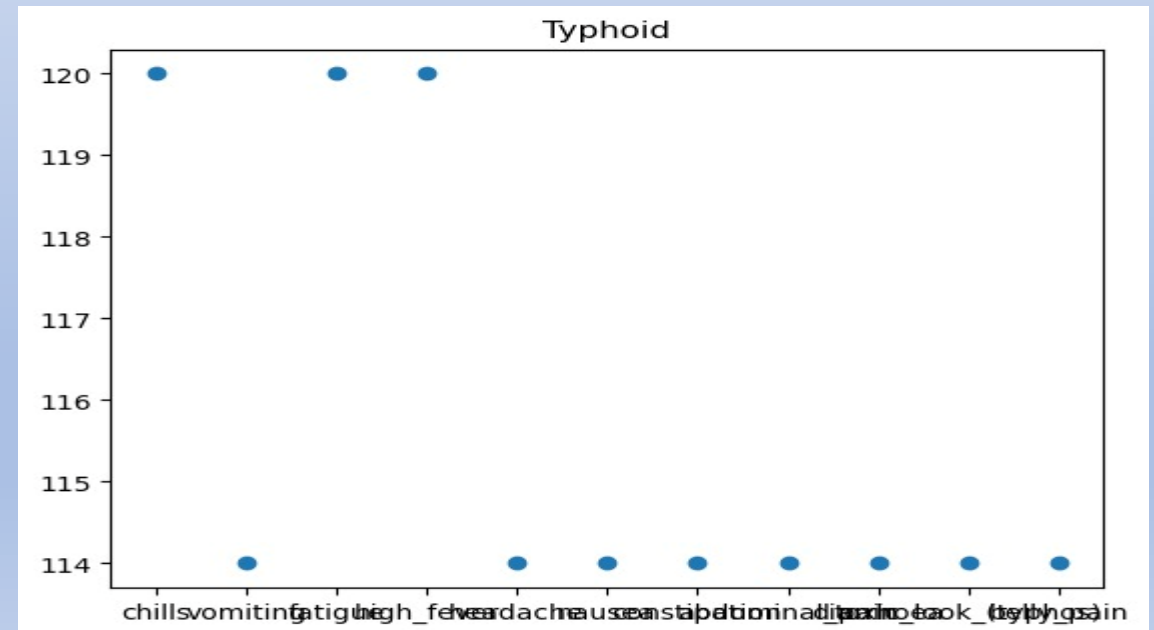
```
Decision Tree
Accuracy
0.9512195121951219
39
Confusion matrix
[[1 0 0 ... 0 0 0]
 [0 1 0 ... 0 0 0]
 [0 0 1 ... 0 0 0]
 ...
 [0 0 0 ... 1 0 0]
 [0 0 0 ... 0 1 0]
 [0 0 0 ... 0 0 1]]
['belly_pain', 'blackheads', 'congestion', 'constipation', 'diarrhoea']
[1, 1, 1, 1, 1]
```





Random Forest: Random Forest is an ensemble learning method that constructs multiple decision trees during training. Each tree is built using a random subset of features and a random subset of data samples, reducing overfitting and increasing robustness. The final prediction is made by aggregating the predictions of individual trees, resulting in a more accurate and stable model for classification and regression tasks.

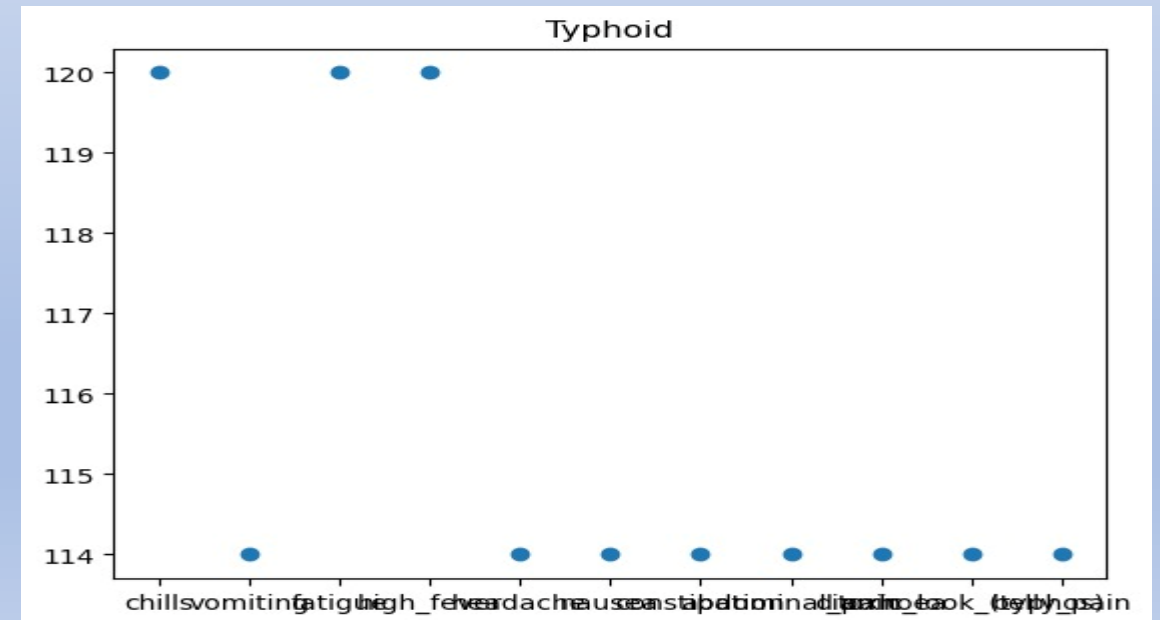
```
Random Forest
Accuracy
0.9512195121951219
39
Confusion matrix
[[1 0 0 ... 0 0 0]
 [0 1 0 ... 0 0 0]
 [0 0 1 ... 0 0 0]
 ...
 [0 0 0 ... 1 0 0]
 [0 0 0 ... 0 1 0]
 [0 0 0 ... 0 0 1]]
[120 114 120 120 114 114 114 114 114 114]
11
11
```





Naive Bayes : is a probabilistic machine learning algorithm based on Bayes' theorem with the assumption of independence between features. It's efficient for classification tasks, especially with high-dimensional data, as it requires a small amount of training data to estimate parameters. Despite its simplicity, Naive Bayes often performs well and is widely used in text classification, spam filtering, and recommendation systems.

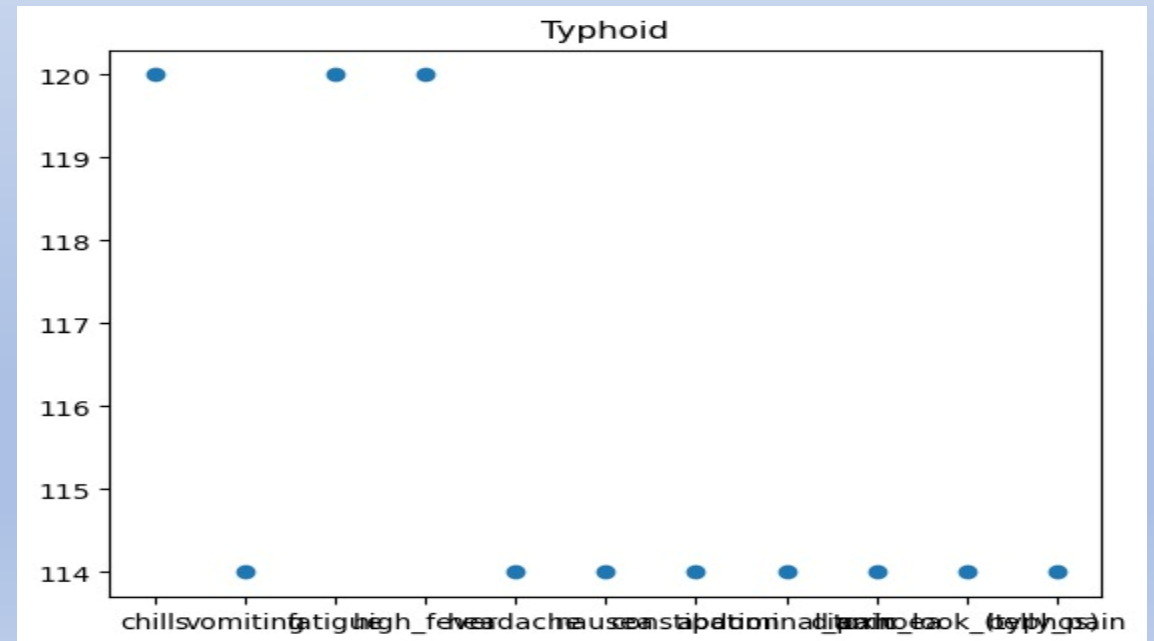
```
Naive Bayes
Accuracy
0.9512195121951219
39
Confusion matrix
[[1 0 0 ... 0 0 0]
 [0 1 0 ... 0 0 0]
 [0 0 1 ... 0 0 0]
 ...
 [0 0 0 ... 1 0 0]
 [0 0 0 ... 0 1 0]
 [0 0 0 ... 0 0 1]]
[120 114 120 120 114 114 114 114 114 114 114]
11
11
```





KNN: is a supervised learning algorithm. It is a basic yet essential algorithm. It finds extensive use in pattern finding and data mining. It works by finding a pattern in data which links data to results and it improves upon the pattern recognition with every iteration. We have used K Nearest Neighbor to classify our dataset and achieved ~95% accuracy

```
kNearest Neighbour
Accuracy
0.9512195121951219
39
Confusion matrix
[[1 0 0 ... 0 0 0]
 [0 1 0 ... 0 0 0]
 [0 0 0 ... 0 0 0]
 ...
 [0 0 0 ... 1 0 0]
 [0 0 0 ... 0 1 0]
 [0 0 0 ... 0 0 1]]
[120 114 120 120 114 114 114 114 114 114 114]
11
11
```





OUTPUT:

Smart Health Care Diagnosis

Smart Health Care Diagnosis Using Machine Learning

Done By: Akhila, Gokul, Shannu, Surya

Name of the Patient *

Symptom 1 *

Symptom 2 *

Symptom 3

Symptom 4

Symptom 5

Prediction 1

Prediction 2

Prediction 3

Prediction 4

Reset Inputs

Exit System

DecisionTree

RandomForest

NaiveBayes

kNearestNeighbour

Typhoid

Typhoid

Typhoid

Typhoid



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A" Grade



FUTURE SCOPE:

- **Advanced Algorithm Integration:** Exploring deep learning and ensemble methods for improved prediction accuracy.
- **Multi-Modal Data Incorporation:** Extending to genetic, imaging, and omics data for comprehensive prediction.
- **Real-Time Monitoring and Telemedicine:** Enabling remote consultations and proactive disease monitoring.
- **Explainable AI Implementation:** Ensuring transparency and trust through interpretable predictions.
- **Global Deployment and Accessibility:** Addressing barriers for widespread access and utilization worldwide.



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A" Grade



CONCLUSION:

In summary, the integration of KNN, Naive Bayes, Random Forest, and Decision Trees in our smart disease prediction system showcases remarkable potential for revolutionizing healthcare diagnostics. Through robust analysis of patient symptom data, these algorithms offer accurate and timely predictions, facilitating early disease detection and intervention. By empowering healthcare professionals with advanced predictive insights, our system holds promise for improving patient outcomes and optimizing resource allocation. This project marks a significant step towards enhancing healthcare efficiency and accessibility, paving the way for a more effective and patient-centric approach to disease diagnosis.

THANK

you